SL Math School: Random Growth Models, etc.
Tutorial IV: Do You Know Brownian Motion?

1 Core Problem

Problem 1 (Tetris)**.** Fix $T > 0$ and define

$$g(x) = T^2 - x^2,$$
$$f(x) = x^2.$$

Let $B : [-T, T] \to \mathbb{R}$ be a Brownian bridge with endpoints $B(-T) = B(T) = 0$. Let $W : [-T, T] \to \mathbb{R}$ be a Brownian bridge with endpoints $W(-T) = W(T) = 0$, conditioned to satisfy

$$W(x) > g(x) \qquad \text{for all } x \in (-T, T).$$

This conditioning is on a probability zero event, so it may be interpreted as a limiting conditioning where the endpoints are raised slightly above the barrier.

Fix $0 < t < T$. Let $\mathbb{P}_t$ denote the law of B restricted to $[-t, t]$, and let $\tilde{\mathbb{P}}_t$ denote the law of W restricted to $[-t, t]$. For a path $X : [-t, t] \to \mathbb{R}$, define

$$\hat{X}(x) = X(x) - X(-t), \qquad x \in [-t, t],$$

and define

$$\hat{f}(x) = f(x) - f(-t) = x^2 - t^2.$$

For $\epsilon > 0$, define the event

$$A_\epsilon(X) = \left\{ |\hat{X}(x) - \hat{f}(x)| \le \epsilon \text{ for all } x \in [-t, t] \right\}.$$

(a) Show that there exist constants $C, c > 0$ such that, for all $0 < \epsilon < 1$ small enough,

$$\frac{\tilde{\mathbb{P}}_t(A_\epsilon(W))}{\mathbb{P}_t(A_\epsilon(B))} \ge C \exp\left(-c \frac{(T^2 + t^2)^2}{T - t} \right).$$

(b) Can you guess a general statement for arbitrary functions f and g?

(c) Optional extension: Consider the law of W further conditioned on the event $A_\epsilon(W)$. Using a Gaussian cost or Cameron–Martin type heuristic, predict the typical size of the endpoint heights $W(-t)$ and $W(t)$.

1.0.1 Extra Challenge Problems

Problem 2 (The cost of a spike)**.** Let B be a Brownian bridge on $[-1, 1]$ with endpoints

$$B(-1) = B(1) = 1.$$

For $M > 0$, define

$$f_M(t) = \begin{cases} M(1 - 2|t|), & |t| \le \frac{1}{2}, \\ 0, & \text{otherwise}. \end{cases}$$

Use the Cameron-Martin theorem for piecewise linear functions to show that

$$\lim_{M\to\infty} \frac{1}{M^2} \log \mathbb{P}\left(B(t) \geq f_M(t) \text{ for all } t \in [-1,1]\right) = -1.$$

Hint: Instead of applying the Cameron-Martin theorem to the function f_M, pick a different piece-wise linear function h satisfying

$$h(-1) = h(1) = 0, \qquad h(t) \geq f_M(t).$$

Problem 3 (The typical shape of a conditioned bridge)**.** In the previous problem, you found a piecewise linear function h which gave the best lower bound on the exponential scale for the event

$$B(t) \geq f_M(t) \qquad \text{for all } t \in [-1,1].$$

What does this suggest about the typical shape of the Brownian bridge conditioned on this event?

More generally, suppose $f_M(t) = Mf(t)$ for some fixed nonnegative function f. What variational problem would you expect to determine the typical macroscopic shape of a Brownian bridge conditioned to stay above f_M?